

Climate 5000 Series - Gen 4

12K BTU Single Zone Regular System

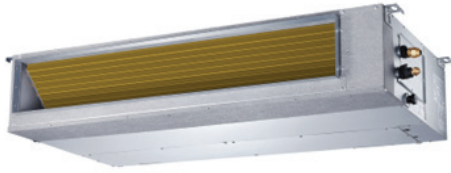
Ducted IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

12K Btu Single Zone Regular System - Ducted IDU

Indoor part number		8-733-962-710
Indoor model number		BMS500-AAU012-1AHDXD
Outdoor part number		8-733-962-685
Outdoor model number		BMS500-AAS012-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		19.0
EER2		11.7
HSPF2 Region IV		10.2
HSPF2 Region V		8.0
COP(47F)		3.42
COP (5F)		2.25
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3100~13000
	Rated Capacity	11500
	Current	4.5
	Input Power	983
Heating	Capacity Range	3700~15600
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	8800
	Current	4.7
	Input Power	1028
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/out-door (A)		3.0 / 12.0
Max. fuse indoor/outdoor (A)		15.0 / 15.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 310
Input (W)		325 / 826
Rated current (RLA/A)		2.40 / 5.65
Piping		
Refrigerant precharge (oz/kg)		32.45 / 0.92
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		82.00 / 25
Max. difference in height (ft/m)		49.20 / 15

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Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.18
IDU	Fan motor model	ZKFN-81-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	150.0
	Fan speed, H / M / L (RPM)	1070 / 990 / 910
	Air flow, Turbo / H / M / L / Si (CFM)	400.2 / 323.7 / 264.9 / 176.6 / 117.7
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 37 / 35 / 33.5 / 26
	Net weight (lbs / kg)	39.46 / 17.9
	Gross weight (lbs / kg)	46.74 / 21.2
ODU	ESP range (Pa)	0-100
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed, H / / L (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1236.1
	Noise level, Max (dB(A))	55.5
	Net weight (lbs / kg)	62.17 / 28.2
	Gross weight (lbs / kg)	67.68 / 30.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Bosch Thermotechnology Corp.
Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

Tel: 1-800-283-3787 www.bosch-homecomfort.us

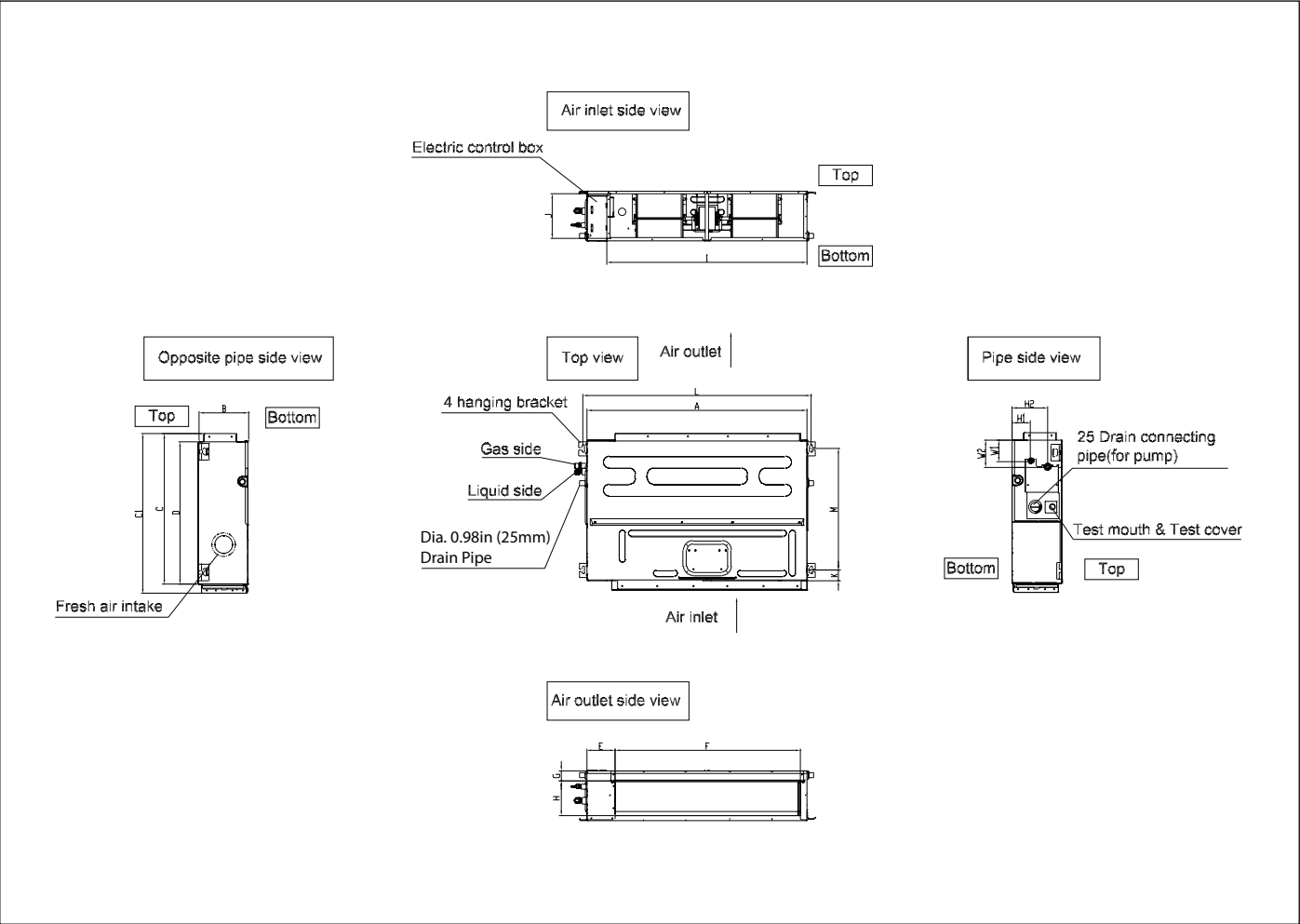
Bosch Thermotechnology Corp. reserves the right to make changes without notice due to continuing engineering and technological advances | BTC 769208350 B | 01.2026

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12K Btu Single Zone Regular System - Ducted IDU



Indoor Unit Dimensions and Connections

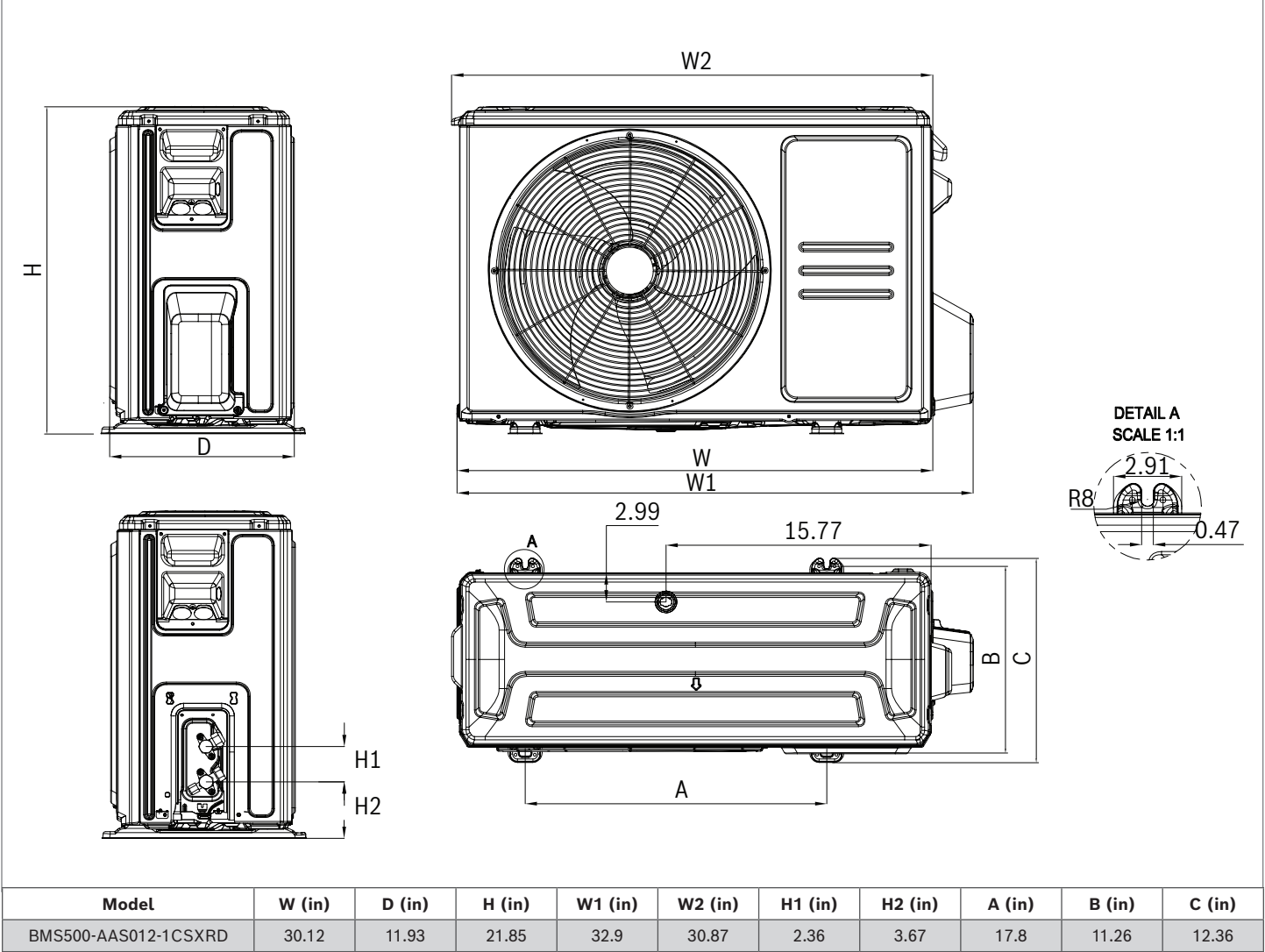


Model	Outline dimension (in)				Air outlet dimension (in)				Return air (in)			Mounting (in)		Gas and liquid valve (in)			
	A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
BMS500-AAU012-1AHDXD	27.6	7.9	19.9	17.7	5.4	21.1	1.2	6.0	23.6	7.3	2.0	29.2	14.2	3.3	5.5	3.3	3.3

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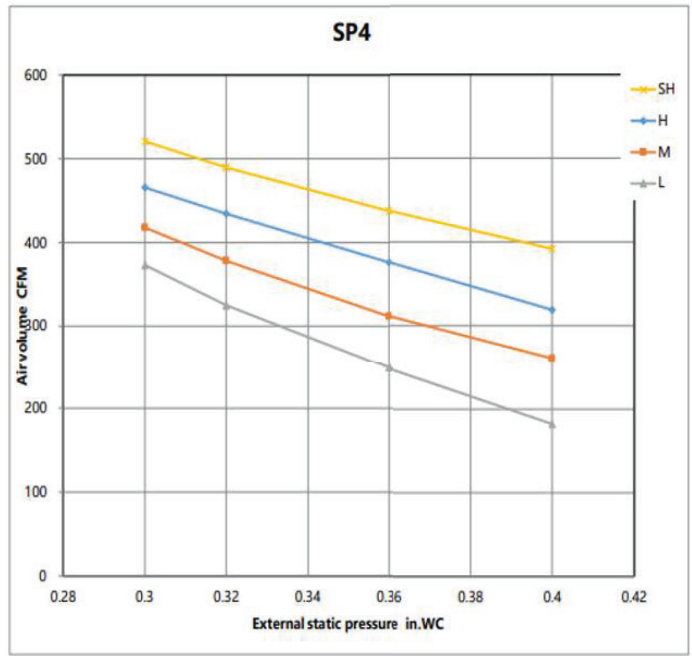
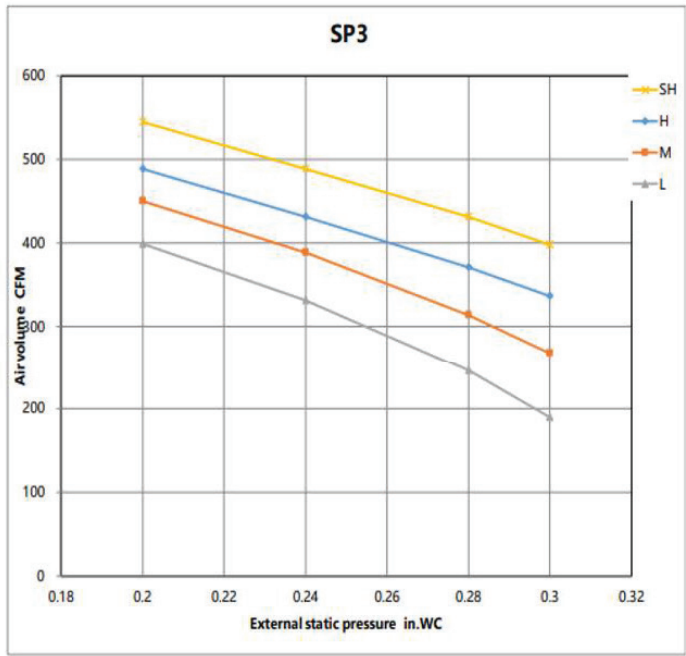
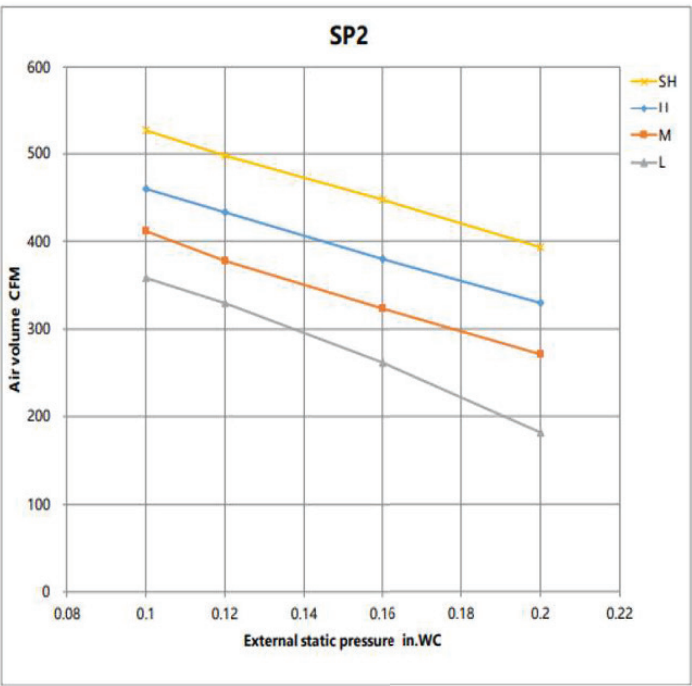
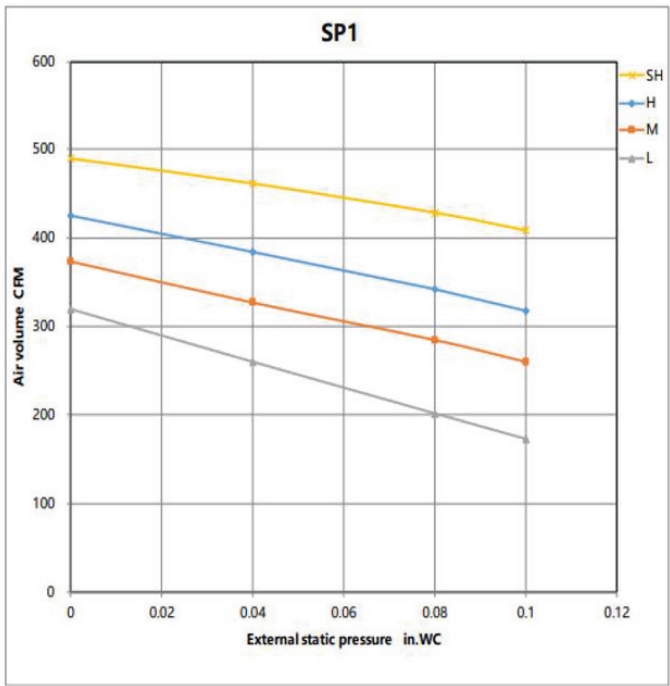


Outdoor Unit Dimensions and Connections

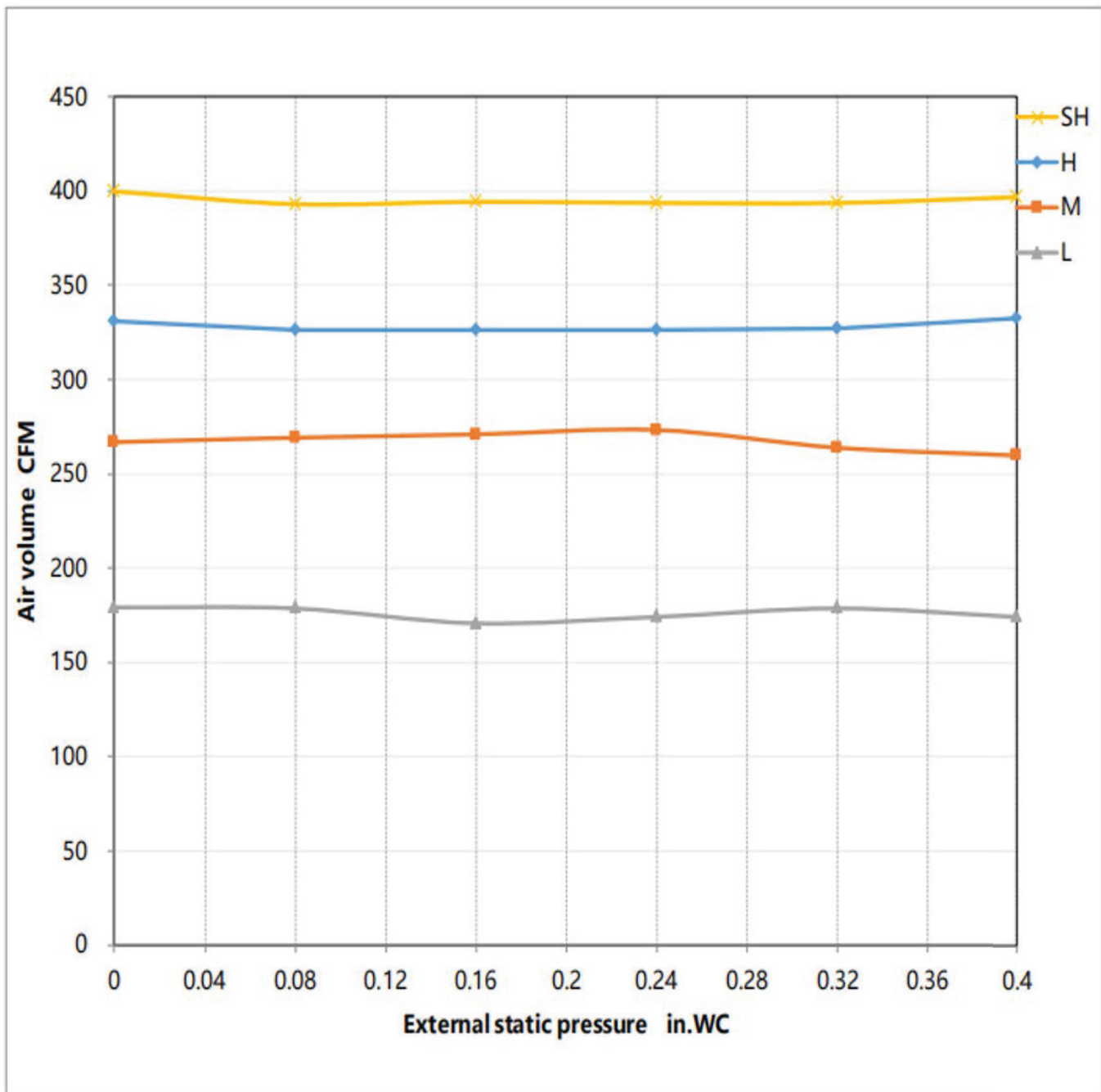




Air Volume



Air Volume



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Cooling Performance - Indoor Dry Bulb Temperature at 400 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	9780	9775	9915	9956	10230	10204	10195	10189	10188	10180	10509	10496	10486	10484	10474	11085	11082	11059	11055	11032
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
-4°F (-20.0°C)	TC	10532	10527	10677	10722	11017	10989	10980	10973	10972	10963	11317	11303	11292	11291	11280	11938	11934	11910	11905	11880
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.63	0.63	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.63	0.63	0.63
0°F (-17.8°C)	TC	10882	10875	11077	11123	11429	11348	11336	11327	11326	11440	11682	11665	11651	11649	11644	12310	12307	12306	12308	12316
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.64	0.64	0.64	0.64	0.64
5°F (-15.0°C)	TC	11194	11232	11489	11542	11898	11719	11722	11722	11734	11909	12093	12096	12099	12099	12115	12803	12799	12806	12807	12813
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.66	0.66	0.66	0.66	0.66
14°F (-10.0°C)	TC	12630	12624	12882	12942	13343	13257	13246	13237	13235	13359	13712	13694	13680	13678	13664	14580	14575	14547	14542	14511
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
17°F (-8.3°C)	TC	13077	13120	13423	13488	13927	13728	13734	13736	13748	13942	14197	14202	14208	14208	14226	15098	15094	15092	15095	15107
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.92	0.92	0.92	0.92	0.92	0.92	0.92
23°F (-5.0°C)	TC	12889	12945	13280	13345	13785	13542	13546	13548	13565	13800	14009	14014	14018	14018	14041	14896	14891	14902	14904	14913
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.87	0.87	0.87	0.87	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.90	0.90	0.90	0.90	0.90
35°F (1.7°C)	TC	12633	12634	12708	12744	13155	13224	13223	13225	13225	13260	13638	13637	13638	13638	13639	14406	14402	14402	14401	14402
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.74	0.74	0.74	0.74	0.78	0.78	0.78	0.78	0.78	0.78	0.81	0.81	0.81	0.82	0.82	0.89	0.89	0.89	0.89	0.89
41°F (5.0°C)	TC	12331	12331	12429	12516	12886	12904	12904	12905	12905	12950	13307	13306	13307	13307	13308	14054	14050	14049	14049	14050
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.59	0.59	0.60	0.61	0.63	0.63	0.63	0.63	0.63	0.63	0.66	0.66	0.66	0.66	0.66	0.72	0.72	0.72	0.72	0.72
47°F (8.3°C)	TC	12028	12036	12197	12251	12614	12585	12585	12585	12585	12643	12976	12975	12976	12976	12976	13703	13699	13698	13697	13699
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.45	0.45	0.46	0.46	0.48	0.48	0.48	0.48	0.48	0.49	0.51	0.50	0.50	0.51	0.51	0.55	0.55	0.55	0.55	0.55
50°F (10.0°C)	TC	11599	11918	12321	12392	12872	12254	12261	12345	12384	12885	12732	12749	12755	12756	12879	13610	13645	13681	13685	13702
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
59°F (15.0°C)	TC	10962	11347	11750	11821	12302	11599	11615	11746	11834	12314	12077	12089	12094	12111	12324	12968	12994	13020	13023	13034
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.54	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
68°F (20.0°C)	TC	10325	10765	11165	11236	11712	10930	10955	11177	11248	11724	11403	11411	11432	11442	11734	12387	12309	12329	12331	12337
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.57	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.59	0.59	0.59	0.59	0.59
77°F (25.0°C)	TC	9683	10150	10547	10617	11092	10266	10298	10558	10629	11103	10731	10743	10770	10782	11112	11590	11625	11652	11654	11665
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.61	0.61	0.61	0.61	0.62	0.61	0.61	0.61	0.61	0.62	0.62	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62	0.62
82°F (27.8°C)	TC	9752	10249	10663	10735	11227	10334	10374	10672	10747	11238	10814	10828	10862	10875	11248	11693	11737	11767	11770	11783
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.69	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.71	0.71	0.71	0.71	0.71	0.71
86°F (30.0°C)	TC	10374	10892	11338	11417	11948	10941	10991	11350	11430	11961	11459	11472	11514	11531	11971	12406	12454	12483	12486	12498
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.85	0.85	0.86	0.86	0.86	0.85	0.85	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.87	0.87	0.87	0.87	0.87
95°F (35.0°C)	TC	11703	11767	12147	12219	12702	12438	12445	12449	12464	12718	12972	12975	12983	12984	13005	13978	13973	13975	13979	13988
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	1.15	1.15	1.17	1.18	1.20	1.19	1.19	1.19	1.19	1.20	1.21	1.21	1.21	1.21	1.22	1.26	1.26	1.26	1.26	1.26
104°F (40.0°C)	TC	10530	10626	11026	11093	11535	11193	11200	11220	11232	11549	11671	11676	11682	11682	11716	12580	1			

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Heating Performance - Indoor Dry Bulb Temperature at 400 CFM

Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	5865	5747	5719	5696	5644	0.95	0.96	0.96	0.96	0.96
-4°F(-20.0°C)	7075	6934	6899	6871	6810	1.03	1.03	1.04	1.04	1.04
0°F(-17.8°C)	7941	7783	7744	7713	7643	1.07	1.08	1.08	1.09	1.09
5°F(-15.0°C)	9024	8844	8800	8764	8685	1.13	1.14	1.15	1.15	1.15
14°F(-10.0°C)	10203	9999	9949	9909	9820	1.15	1.16	1.16	1.17	1.17
17°F(-8.3°C)	10595	10383	10332	10290	10198	1.16	1.17	1.17	1.17	1.18
22°F(-5.6°C)	10836	10620	10567	10524	10430	1.16	1.17	1.17	1.17	1.17
27°F(-2.8°C)	11077	10856	10802	10758	10662	1.16	1.16	1.17	1.17	1.17
32°F(-0.0°C)	11319	11092	11037	10993	10894	1.15	1.16	1.17	1.17	1.17
35°F(1.7°C)	11463	11234	11178	11133	11033	1.15	1.16	1.17	1.17	1.17
37°F(2.8°C)	12225	11980	11920	11873	11766	1.18	1.19	1.19	1.19	1.20
42°F(5.6°C)	14129	13846	13777	13722	13598	1.24	1.25	1.25	1.25	1.26
44.6°F(7.0°C)	14891	14593	14520	14462	14332	1.26	1.27	1.27	1.27	1.28
47°F(8.3°C)	16033	15712	15634	15571	15431	1.29	1.31	1.31	1.31	1.32
52°F(11.1°C)	15441	15132	15056	14996	14861	1.15	1.16	1.17	1.17	1.17
57°F(13.9°C)	14849	14552	14479	14421	14291	1.01	1.02	1.02	1.03	1.03
59°F(15.0°C)	14612	14319	14248	14191	14063	0.96	0.97	0.97	0.97	0.97
62°F(16.7°C)	14256	13971	13901	13846	13721	0.87	0.88	0.88	0.88	0.89
64.4°F(18.0°C)	13972	13693	13624	13570	13447	0.81	0.81	0.81	0.82	0.82
69.8°F(21.0°C)	13717	13442	13375	13322	13202	0.65	0.66	0.66	0.66	0.66
75.2°F(24.0°C)	10228	10023	9973	9933	9844	0.50	0.51	0.51	0.51	0.51

Bolded values were tested. All other values are simulated and for reference only.